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Assessing Monetary Policy Globally: Evidence from LLMs and Semi-structural Models

WanTing Xu and Diego Vilán

Over the past decades, central bank communication has become an increasingly important component of modern monetary policy frameworks. Policy rates, asset purchases, and balance-sheet tools remain central to the transmission of monetary policy, but speeches, press conferences, forward guidance, and other forms of public communication have all played a direct role in shaping expectations about the future path of policy. This is particularly important in a global setting, where signals from major central banks influence exchange rates, capital flows, risk appetite, and financial conditions well beyond national borders. Assessing monetary policy developments globally therefore requires not only tracking observed policy actions but also interpreting how central banks communicate their views on inflation, growth, financial stability, and the likely direction of future policy.

Traditional macroeconomic and semi-structural models provide a disciplined approach for evaluating monetary conditions. These models typically summarize the policy stance through observable macro-financial variables, estimated reaction functions, output gaps, inflation pressures, and assumptions about monetary transmission. Their strength lies in their economic structure and interpretability. However, they are often constrained by data lags, revisions, country-specific modeling choices, and the difficulty of incorporating unstructured textual information. Central bank speeches contain timely and forward-looking information, much of which is qualitative, contextual, and difficult to summarize using standard quantitative indicators alone.

Recent advances in artificial intelligence offer new ways to extract information from policy communication. Large language models (LLMs) can process large volumes of text consistently across countries and over time, while capturing nuance, context, conditionality, and tone in ways that simple keyword-based approaches may miss. This is especially valuable for central bank communication, where the same word may carry different policy implications depending on the surrounding discussion of inflation, financial conditions, uncertainty, or future risks. By translating speeches into structured sentiment measures, language-model-based methods can provide a timely textual signal of the intended direction of monetary policy.

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In this note, we apply these methods to a cross-country collection of central bank speeches from the Bank for International Settlements (BIS) database[1]. We construct two monetary-policy sentiment measures using a traditional lexicon-based approach and an LLM-based approach, allowing us to compare whether model-based interpretation of language produces a materially different signal from dictionary-based scoring. The resulting speech-level measures are then aggregated into country-level and global indicators of monetary policy communication. Because countries with more frequent central bank communication naturally receive greater weight in the corpus, the aggregate measure can be interpreted as a communication-weighted global monetary policy tone.

The main objective of this note is to assess whether language-model-based measures of central bank communication provide a useful and timely characterization of the global monetary policy environment, and how these measures compare and relate to more traditional macro semi-structural approaches. In turn, we present a comparative analysis evaluating the interpretations of Large Language Models (LLMs) against traditional semi-structural model approaches. We evaluate whether LLMs can capture, or potentially provide early signals of, shifts in the direction of monetary policy around major episodes such as the Asian Financial Crisis (AFC), the Global Financial Crisis (GFC), the COVID-19 pandemic, and the post-pandemic tightening cycle. The analysis asks whether these textual signals merely reproduce familiar macroeconomic patterns, or whether they contain additional information about the direction, intensity, and synchronization of global monetary policy.

The analysis proceeds in three stages. First, we construct lexicon- and LLM-based measures of the hawkish or dovish tone of individual central bank speeches. We aggregate these scores over time, compare the resulting global indicators, and examine their country-level patterns. We also consider how differences in speech frequency affect the raw global aggregates. Second, we estimate the monetary policy stance for a handful of countries using semi-structural models (SSMs) in the spirit of González-Astudillo and Vilán (2025) and compare these estimates with the policy tone conveyed through central bank communication. Third, we estimate separate dynamic factor models for the LLM- and SSM-based country series to summarize their common variation. We estimate the LLM factor using both the full country sample and a selected eight-country sample, allowing us to assess its sensitivity to country coverage. We then compare the eight-country LLM factor with the SSM factor estimated using the same country set. Together, these exercises allow us to examine how communicated policy guidance relates to the macro-implied stance and whether the two approaches provide complementary perspectives on the evolution of monetary policy globally.

Measures of Global Monetary Policy Tone

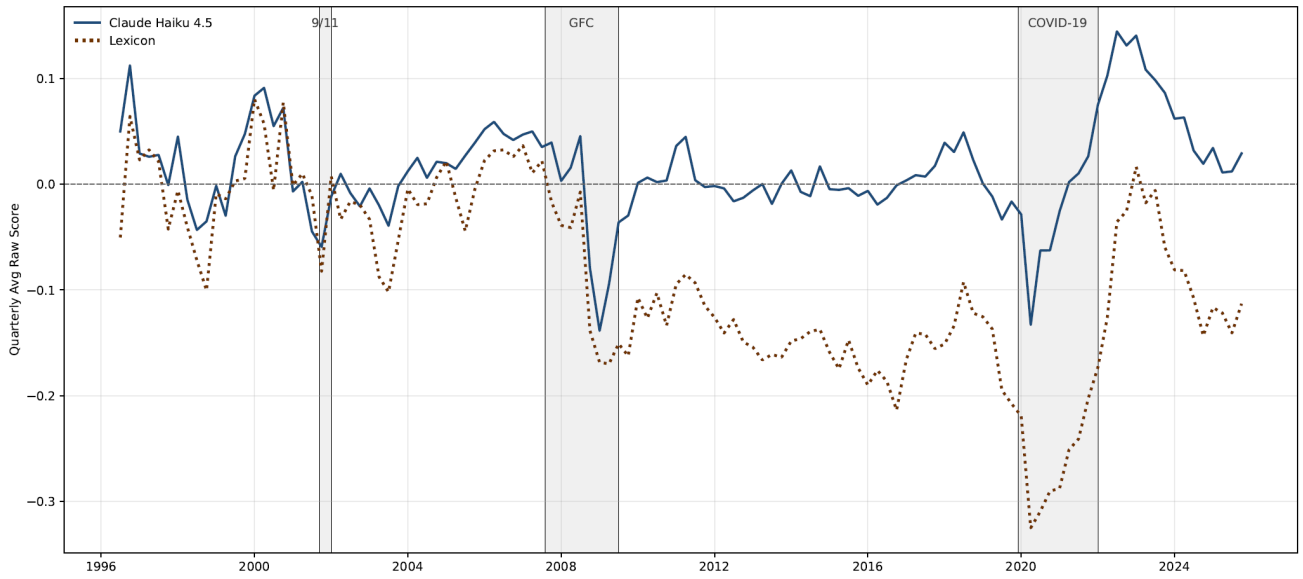
We construct two sentiment measures from the same BIS central bank speech collection, spanning 1996 to 2025. The first is a lexicon-based model that scores each speech using a curated sentiment dictionary comprising hawkish and dovish terms.[2] The second is an LLM-based model that employs Claude Haiku 4.5.[3] Both models produce a speech-level raw score $s \in [-1, 1]$, where $+1$ denotes strongly hawkish, and -1 denotes strongly dovish. To ensure direct comparability, both measures are constructed from the same set of central bank speeches. This

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note contributes to a growing literature that uses artificial intelligence and other text-based methods to measure the tone of monetary policy communication. Rutkowska and Szyszko (2024) compare lexicons for monetary-policy communication, providing a benchmark for our dictionary measure. Hansen and Kazinnik (2024) show that GPT can classify the stance of FOMC announcements, while Peskoff et al. (2023) use GPT-4 to quantify disagreement among FOMC participants. At the global level, Baird et al. (2026) apply LLMs to monetary-policy statements, and Silva, Moriya, and Veyrune (2025) classify topic, stance, sentiment, and audience across 169 central banks. Our note complements this work by comparing a transparent lexicon with a general-purpose LLM on the same speech corpus and benchmarking the resulting global tone against macro-implied policy stances, a comparison motivated by the cross-border transmission documented by Miranda-Agrippino and Rey (2020).

Figure 1 presents the quarterly average sentiment scores derived from the LLM and Lexicon models. Prior to the GFC, both series broadly track each other, reflecting alternating episodes of easing and tightening. Both measures turn dovish during the Asian Financial Crisis, the dot-com bust and the September 11 attacks. During the GFC, both measures drop sharply, consistent with the unprecedented monetary easing adopted by central banks worldwide. In the post-GFC period, however, the two series begin to diverge: the Lexicon measure remains persistently dovish while the LLM measure recovers toward neutral. Despite this divergence, both series still share a common trend, differing primarily in magnitude. During the COVID-19 period, this pattern continues: both measures transition from dovish to hawkish, with the LLM measure becoming increasingly hawkish and the Lexicon measure becoming progressively less dovish. Overall, the two series exhibit a Pearson correlation of 0.49, reflecting co-movement in trend but divergence in magnitude.

Figure 1. Global Monetary Policy Tone. LLM vs. Lexicon-based Measures



Note: Positive scores indicate more hawkish language, while negative scores indicate less hawkish language.
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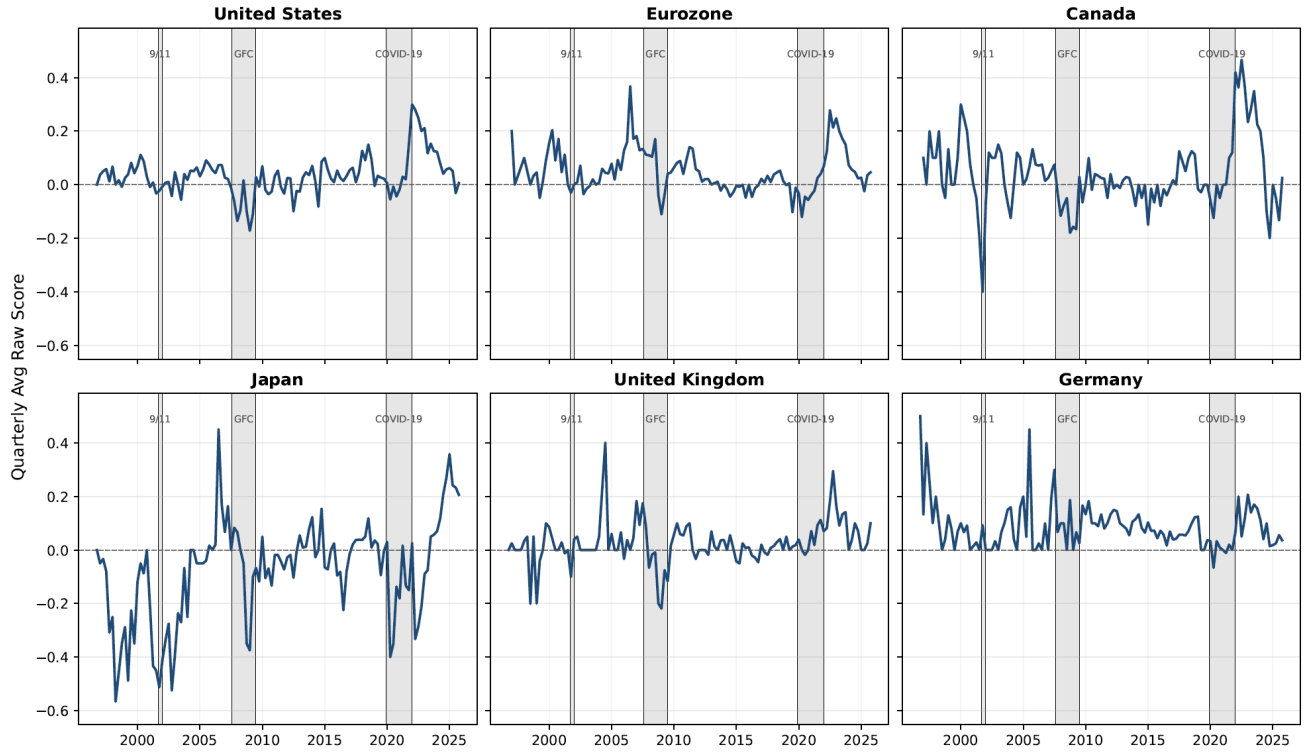
The global quarterly sentiment score shown in Figure 1 represents a speech-weighted measure of global monetary policy tone, not an equal-weighted measure across countries. Consequently, countries with more speeches, predominantly major advanced economies, naturally exert greater influence on the aggregate score. During the AFC, for instance, the aggregate dovish signal primarily reflects the easing adopted by the Federal Reserve and other major central banks, rather than the contractionary policies imposed on crisis-hit Southeast Asian economies, whose speeches are underrepresented in the corpus. This implicit weighting may nevertheless be economically meaningful, since central banks that communicate more frequently tend to be those with greater global influence, and their policy signals may generate broader cross-border spillovers. In this sense, the aggregate score approximates a communication-weighted measure of global monetary policy tone, in which more vocal and systemically important central banks carry greater weight.

To make this implicit weighting explicit, we identify the six countries with the highest speech counts among the 119 countries in our sample. Since these countries contribute the most speeches, they are likely to exert the greatest influence on the global aggregate. We therefore focus on these six countries to examine the communication patterns underlying the LLM-based measure of the direction of monetary policy globally.

Figure 2 presents the country-level patterns over the past three decades. We interpret these scores as measures of the direction of monetary policy conveyed through central bank communication. The country-level series exhibit substantial co-movement around major episodes, including a collective dovish shift around the September 11 attacks and during the GFC, followed by broadly hawkish signals in the post-COVID period. Nonetheless, notable country-specific patterns remain. The Bundesbank series exhibits a relatively hawkish tone throughout much of the sample, consistent with its well-known emphasis on price stability. The model also identifies the Bank of Japan's prolonged dovish tone and a distinct dovish shift in European Central Bank communication during the European sovereign debt crisis, reflecting the extraordinary easing measures adopted during that period.

Figure 2. LLM-Based Monetary Policy Tone: Selected Countries

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The semi-structural benchmark

To benchmark the LLM-derived monetary policy communication factor, we compare it with a semi-structural measure of the monetary policy stance. The model follows the spirit of González-Astudillo and Vilán (2025), who estimate country-specific neutral nominal interest rates using a semi-structural framework that jointly characterizes the trend and cyclical components of output, unemployment, inflation, long-term government bond yields, and the policy interest rate. This approach is particularly useful since it provides a macroeconomic benchmark for the policy stance that is grounded in observable economic conditions, while still allowing key equilibrium variables to vary over time.

The central object of the model is the interest-rate gap, defined as the difference between the policy rate and the country-specific neutral nominal interest rate. For country i at time t, the neutral nominal rate is given by:

$$R^*_{i,t} = r^*_{i,t} + \pi^*_{i,t}$$

where $r^*_{i,t}$ is the natural real rate of interest and $\pi^*_{i,t}$ is trend inflation. The monetary policy stance is then measured as:

$$MPS_{i,t} = R_{i,t} - R^*_{i,t}$$

A positive value of $\mathrm{MPS}_{i,t}$ indicates that the policy rate is above the neutral nominal rate and [illegible] is restrictive. A negative value indicates that the policy rate is below neutral

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and that monetary policy is accommodative. This distinction is important because the same observed policy rate can imply different monetary conditions across countries if their neutral rates differ. González-Astudillo and Vilán emphasize this point in the Eurozone context, where member states share a common ECB policy rate but may face different country-specific stances.

The model consists of three main blocks. The first block of the model describes real activity and inflation. Output is decomposed into potential output and an output gap:

$$y_{i,t} = y^*_{i,t} - c_{i,t}$$

where $y_{i,t}$ is log real output, $y^*_{i,t}$ is potential output, and $c_{i,t}$ is the output gap. The unemployment rate is linked to the output gap through an Okun's-law relationship, while inflation is modeled through a Phillips-curve relationship in which inflation depends on lagged inflation, trend inflation, and the output gap:

$$u_{i,t} = u^*_{i,t} + \theta_{i,1} c_{i,t} + \theta_{i,2} c_{i,t-1} + \nu_{i,t}$$

$$\pi_{i,t} = \beta_i \pi_{i,t-1} + (1 - \beta_i)\pi^*_{i,t} + \kappa_i c_{i,t} + \varepsilon^{\pi}_{i,t}$$

This block gives the model its aggregate demand and aggregate supply structure. The output gap summarizes cyclical demand pressures, while the inflation equation captures the link between slack and price dynamics. In this sense, the model uses macroeconomic fundamentals to infer whether monetary conditions are expansionary or contractionary.

A distinctive feature of González-Astudillo and Vilán's (2025) framework is that the output gap responds to both short-run and long-run interest-rate gaps. The short-run gap is based on the policy or shadow policy rate relative to expected inflation and the natural real rate. The long-run gap is based on country-specific long-term government bond yields. This matters because monetary policy is transmitted not only through overnight or short-term policy rates, but also through the yield curve, sovereign spreads, term premia, and broader financing conditions. The paper explicitly motivates the long-run interest-rate gap as a way to capture sovereign debt risk and other country-specific premia, which can be especially important during periods of financial stress.

The long-term interest-rate block relates the country-specific 10-year government bond yield to the natural real rate, trend inflation, and two additional components:

$$i^{10}_{i,t} = r^*_{i,t} + \pi^*_{i,t} + p^{10}_{i,t} + c^{10}_{i,t}$$

Here, $p^{10}_{i,t}$ captures persistent movements in term or country premia, while $c^{10}_{i,t}$ captures the cyclical long-run real interest-rate gap. This block helps identify the natural rate of interest, especially when the policy rate is constrained by the zero lower bound and therefore provides limited information about the underlying stance of policy.

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The policy-rate block closes the model through an inertial Taylor rule that depends on the natural real rate, trend inflation, inflation deviations from target, and the output gap. Following González-Astudillo and Vilán (2025), the observed policy rate is treated as a censored realization of an underlying shadow rate. When the zero lower bound binds, the interest-rate gap is computed as the model-implied shadow rate minus the estimated neutral nominal rate. The estimated stance can therefore vary even when the observed policy rate remains unchanged at zero, allowing the model to capture, at least partially, accommodation associated with unconventional monetary policy.

The model's latent variables (potential output, the natural unemployment rate, trend inflation, and the natural real interest rate) are unobserved series inferred jointly within this framework. Bayesian estimation combines information from observed macroeconomic and financial data with the model's economic relationships and prior distributions. Output, unemployment, and inflation help distinguish persistent movements in equilibrium conditions from cyclical fluctuations, while the policy-rate and long-term yield equations provide additional information about the natural real rate. These relationships jointly discipline the estimates of the latent states. Furthermore, Bayesian estimation characterizes uncertainty about the latent states and model parameters.

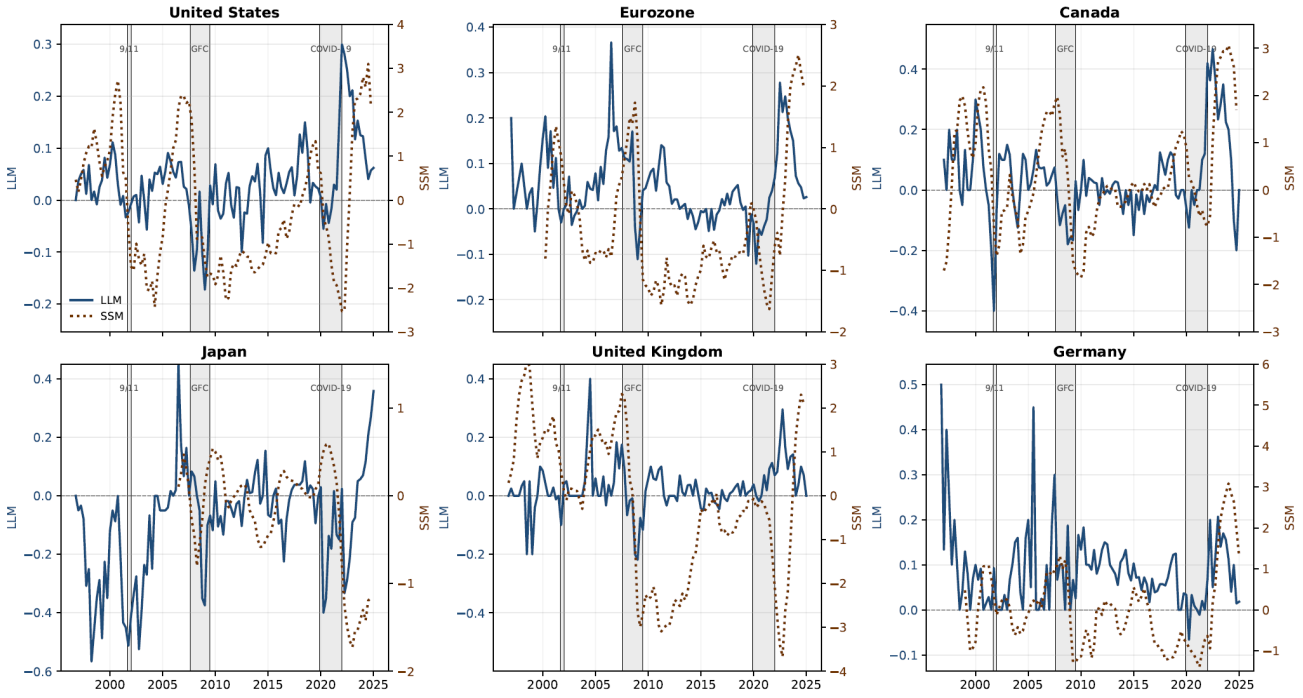
For our analysis, the semi-structural model serves as the macroeconomic benchmark against which the LLM-based communication factor is evaluated. The two measures are conceptually different. The semi-structural model infers the stance from macroeconomic data, including output, unemployment, inflation, policy rates, and long-term yields. The LLM-based measure infers the tone from central bank communication. Agreement between the two measures would suggest that central bank speeches largely reflect the contemporaneous macroeconomic stance implied by fundamentals. Divergence between them may be informative, as it could capture forward guidance, risk-management concerns, financial stability considerations, or early communication about future policy turning points before those shifts are fully visible in realized macroeconomic data.

Results

We interpret the semi-structural interest-rate gap as a benchmark measure of the macro-implied monetary policy stance, and the LLM factor as a measure of the policy orientation conveyed through central bank communication. The comparison allows us to assess whether central bank language provides incremental information beyond traditional macroeconomic indicators, and whether that information is especially useful around major global policy turning points such as the Global Financial Crisis, the COVID-19 pandemic, and the post-pandemic tightening cycle. Figure 3 displays the LLM and SSM results for six representative economies. The two measures are expressed in different units, so the comparison emphasizes the direction, timing, and persistence of their movements rather than their absolute magnitudes.

Figure 3. Monetary Policy Tone and Stance by Country

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Across the displayed economies, co-movement is most apparent around major global policy turning points. The communication measure generally changes more decisively, while the semi-structural measure adjusts more gradually and remains persistent for longer. Around the Global Financial Crisis, the LLM measure turns dovish rapidly as central banks communicate emergency support and easing, whereas the SSM measure often continues to reflect the restrictive conditions that preceded the crisis before moving into accommodative territory. A similar sequence appears around the COVID-19 shock and the subsequent inflation episode: communication first shifts toward support and then toward tightening, while the macro-implied stance responds later as policy rates, inflation, output, and long-term yields incorporate the new environment.

The clearest post-pandemic sequencing appears in Canada, the Eurozone, the United Kingdom, and the United States. In each case, the LLM series begins to move in a hawkish direction before the SSM gap reaches its restrictive peak. The subsequent moderation in the LLM series also occurs while the SSM measure remains elevated, consistent with central banks communicating an anticipated change in the direction of policy before the estimated macroeconomic stance has fully adjusted. Around the Global Financial Crisis, by contrast, both measures ultimately signal accommodation, but the SSM response is generally smoother and more prolonged. These differences are consistent with the SSM's dependence on persistent latent variables, including the natural rate and trend inflation, and with the faster-moving nature of policy communication.

Taken together, Figure 3 points to complementarity rather than interchangeability between the two measures. The LLM series appears to provide a timelier indicator of changes in the direction of policy, while the SSM supplies a slower-moving benchmark anchored in macroeconomic

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fundamentals. This pattern is consistent with the forward-looking nature of central bank communication and suggests that the LLM measure captures leading information about subsequent changes in monetary policy stance, although the timing and magnitude of this lead vary across countries and episodes.

The Dynamic Factor Model

Building on the country-level results, we posit a single latent-factor structure for each measure to capture common variation across countries. We interpret the LLM factor as capturing the common monetary policy tone conveyed through central bank communication and the SSM factor as capturing the common variation in the monetary policy stance. For the factor analysis, we expand the six-country sample shown in Figure 3 to seven countries by adding Australia. Together, these seven countries account for nearly fifty percent of all speeches in the BIS sample. Using this common country set, we estimate separate LLM- and SSM-based dynamic factor models via maximum likelihood. The models share the following state-space structure:

Observation equation:

$$x_{i,t} = \lambda_i f_t + \varepsilon_{i,t}, \quad \varepsilon_{i,t} \sim N\left(0, \sigma_i^2\right)$$

State equation:

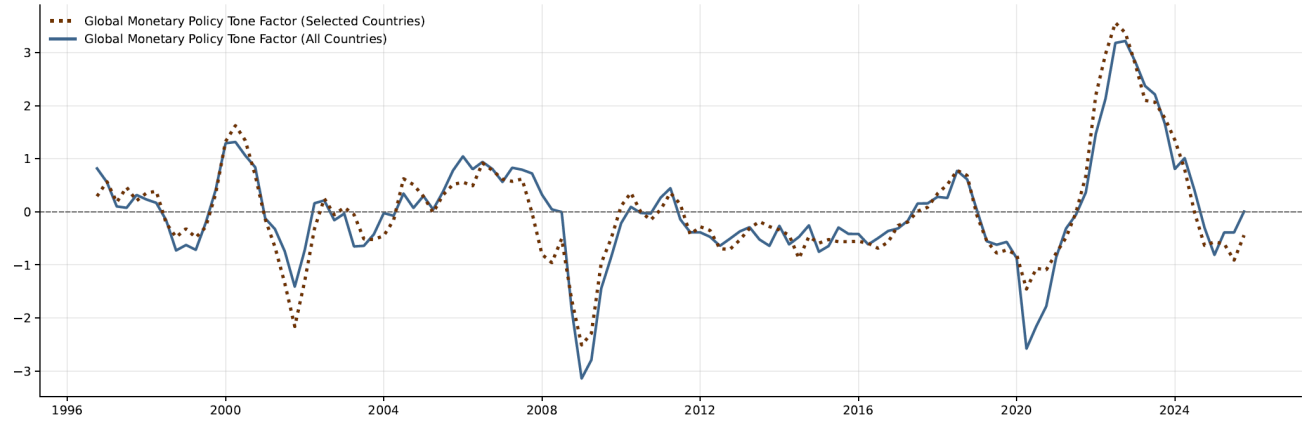
$$f_t = \phi_1 f_{t-1} + \phi_2 f_{t-2} + \eta_t, \quad \eta_t \sim N(0, 1)$$

where $x_{i,t}$ is the standardized country-level observation for country i at time t. In the LLM model, $x_{i,t}$ represents the quarterly average monetary policy tone score; in the SSM model, it is the estimated interest-rate gap. The variable f_t is the corresponding latent common factor, λ_i is the factor loading for country i, $\varepsilon_{i,t}$ is the country-specific idiosyncratic error, and ϕ_1, ϕ_2 are the autoregressive coefficients governing the factor dynamics.

Figure 4 compares the LLM-based communication factor estimated using the full country sample in the BIS database with the corresponding factor estimated using the selected eight-country sample. The two factors co-move closely throughout the sample, particularly around the September 11 attacks, the Global Financial Crisis, and the COVID-19 pandemic. Both shift in a dovish direction around these episodes and subsequently turn toward tightening as economic conditions recover. Their close correspondence suggests that the selected eight-country sample captures much of the common variation in monetary policy communication across countries, although differences in magnitude indicate that it does not reproduce the full-sample factor exactly.

Figure 4. Global Monetary Policy Tone: Full Sample vs Selected Countries

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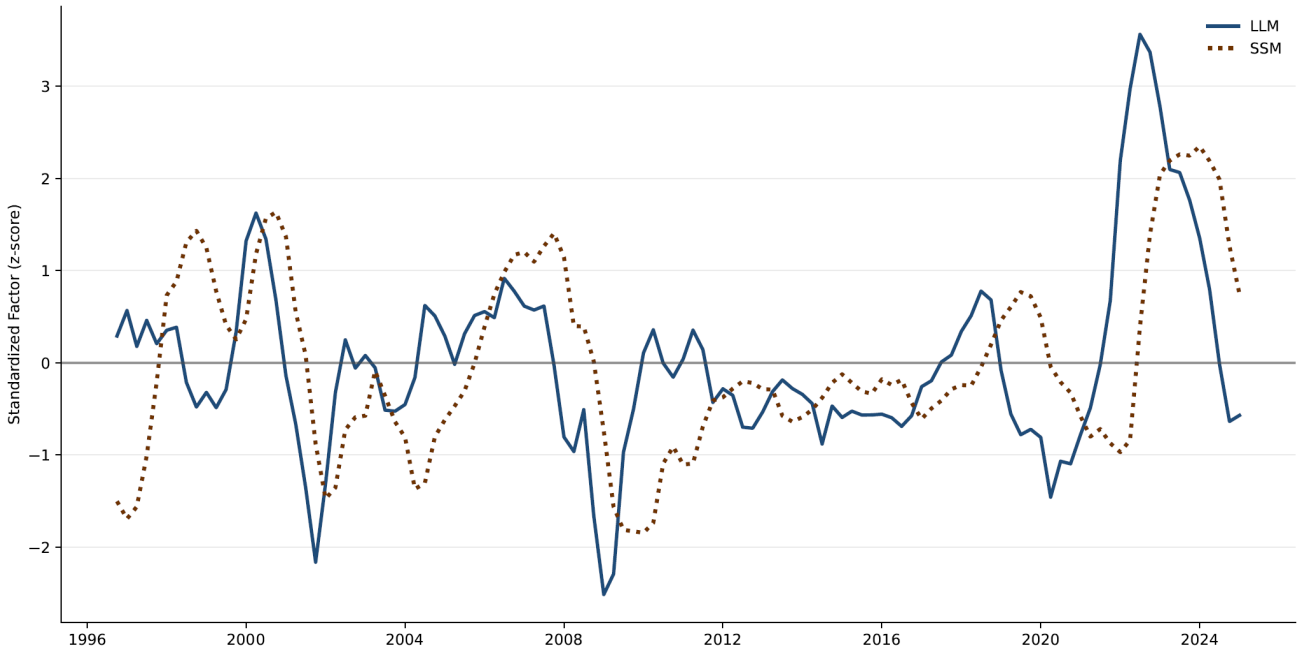
The two factors nevertheless differ somewhat in the magnitude of their movements. During easing episodes, the dovish shifts in the eight-country factor tend to be less pronounced than those in the full-sample factor, whereas during tightening episodes its hawkish shifts are often more pronounced. This pattern may indicate that the most speech-active central banks communicate tightening signals more strongly or explicitly than easing signals. However, because the factors are estimated and standardized separately, differences in magnitude may also reflect sample composition, factor loadings, and country-specific variation. The apparent asymmetry should therefore be interpreted as a descriptive pattern rather than definitive evidence of a systematic signaling bias.

Having examined the sensitivity of the LLM communication factor to country coverage, we next compare it with the common factor extracted from the semi-structural model estimates of the monetary policy stance. To ensure comparability, both factors are estimated using the same eight-country sample and standardized to have a mean of zero and a standard deviation of one. The comparison therefore focuses on whether common movements in communicated policy tone align with those in the policy stance in terms of direction and timing, rather than on differences in their absolute magnitudes.

Figure 5 indicates broad co-movement between the LLM and SSM factors around several major policy turning points, although the two series do not always adjust simultaneously. The LLM factor tends to display greater short-term variation and, in some episodes, appears to turn before the SSM factor, which is smoother and more persistent. This descriptive pattern is consistent with central bank communication responding to changing policy considerations before those developments are fully reflected in the macroeconomic variables informing the SSM. It also suggests that the LLM factor may capture elements of forward guidance contained in central bank speeches, thereby providing an early signal of the future direction of monetary policy.

Figure 5. Global Factors – LLM Policy Tone vs. SSM Policy Stance

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Conclusions

This note presents a comparative analysis evaluating the predictions of LLM models against traditional macro semi-structural model approaches. The first uses central bank speeches to construct lexicon- and LLM-based indicators of the intended direction of monetary policy. The second uses semi-structural models to estimate the position of policy rates relative to country-specific neutral nominal rates. Bringing these measures together provides a way to compare what central banks communicate about the direction of policy with the stance implied by macroeconomic fundamentals.

Three findings stand out. First, the lexicon and LLM measures share a common cyclical component but differ materially in magnitude, especially after the Global Financial Crisis, when the lexicon measure remains persistently dovish while the LLM measure returns closer to neutral. This suggests that the LLM captures context and conditionality that a fixed dictionary does not. Second, the country series and the global communication factor identify synchronized shifts around the September 11 attacks, the Global Financial Crisis, the COVID-19 pandemic, and the post-pandemic tightening cycle. The factor based on the eight most speech-active countries displays particularly pronounced tightening signals, underscoring the influence of systemically important and communication-intensive central banks on the global measure. Third, the LLM and SSM indicators frequently move in the same direction around major turning points, but the communication measure often adjusts earlier and displays greater short-term variation.

These results indicate that the two approaches measure related but different dimensions of monetary policy. The semi-structural framework is disciplined by observed inflation, activity, interest rates, long-term yields, and estimates of equilibrium variables. The LLM measure [illegible] policymakers' interpretation of incoming information, their balance of risks, and their intended policy direction. Temporary divergence between the measures is therefore not

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necessarily a contradiction. It can signal a period in which central banks are preparing markets for a change that has not yet appeared in the macro-implied stance, potentially reflecting a lead-lag relationship. Used together, the indicators may offer a richer assessment than either measure alone: the LLM-based measure captures the timely and forward-looking content of central bank communication, while the SSM supplies economic structure and an interpretable benchmark. In this sense, future research should explore the potential use of LLM-based communication indicators as a timely complement to conventional semi-structural measures of the monetary policy stance.

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Appendix

A. Lexicon Dictionary

Below are the hawkish and dovish terms used by the lexicon model to gauge the monetary policy tone of each speech. The model assigns a speech-level score by comparing language associated with tighter policy with language associated with more accommodative policy.

HAWKISH TERMS (108)

ase, raise rates, raising rates, interest rate hike, policy tightening, tightening monetary policy, tightening of monetary policy, monetary tightening, further tightening, lift-off,

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liftoff, quantitative tightening, balance sheet runoff, reducing accommodation, removing stimulus, tighten policy, 75 basis points, jumbo rate hike, more tightening, inflation not transitory, raise interest rates, increase interest rates, increase in interest rates, firming of policy, policy firming, removing policy accommodation, inflationary pressures, inflationary pressure, sufficiently restrictive, price stability mandate, underlying inflation, persistent inflation, elevated inflation, inflation above target, higher for longer, second-round effects, wage-price spiral, entrenched inflation, inflation remains elevated, de-anchoring, upward pressure on prices, supply bottlenecks, upside risks to inflation, monetary restraint, excess demand, overheating, overheat, restrictive stance, restrictive territory, restrictive policy, tight labour market, tight labor market, nairu, above 2 percent, front-loading, not yet done, upward adjustment, meet our inflation target, wage pressures, wage inflation, above potential, closing output gap, risks are asymmetric, skewed to the upside, preemptive, preemptive action, irrational exuberance, labor market tightness, measured pace, measured tightening, tapering, taper, exit strategy, policy normalization, balance sheet normalization, inflation objective, unit labour costs, unit labor costs, capacity constraints, price pressures, asset price inflation, non-accelerating, convergence criteria, emu convergence, policy accommodation, housing market imbalance, asset price misalignment, unwinding, gradual normalization, tightening, price stability, high inflation, normalisation, normalization, cost pressures, increasing cost, firming, tighten, inflation target, non-inflationary, stability and growth pact, fiscal consolidation, fiscal discipline, upside risks, vigilant, inflation expectations well anchored, global imbalances

DOVISH TERMS (136)

rate cut, rate reduction, cut rates, cut interest rates, interest rate cut, lower rates, monetary easing, easing monetary policy, further easing, policy easing, quantitative easing, asset purchases, asset purchase programme, purchase programme, purchase program, accommodative monetary policy, accommodative stance, monetary accommodation, stimulus, effective lower bound, zero lower bound, interest rate reductions, easy stance, negative interest rates, low for long, whatever it takes, easing cycle, cutting cycle, lower interest rates, reduce interest rates, interest rate cuts, cut the rate, asian crisis, asian financial crisis, russian default, deflationary spiral, zero interest rate, economic stimulus package, subprime crisis, emergency liquidity, sovereign debt crisis, outright monetary transactions, yield curve control, negative policy rate, pandemic, covid, covid-19, coronavirus, recession, global financial crisis, unprecedented shock, support the economy, support growth, credit crunch, unconventional measures, forward guidance, secular stagnation, fiscal stimulus, pivot, disinflation, disinflationary, deflation, deflationary, deflationary pressures, slowdown, downturn, contraction, economic slowdown, downside risks, downside risks to growth, weak demand, economic slack, slack in the economy, high unemployment, job losses, risk of recession, fragile recovery, below 2 percent, low interest rates, ease policy, scope to ease, inject liquidity, growth slowdown, labor market weakness, labour market weakness, rising unemployment, external shock, currency crisis, contagion, capital outflows, speculative attack, financial turmoil, russian crisis, ltcm, lost decade, balance sheet recession, insurance cut, mid-cycle adjustment, stimulus package, subprime, housing bubble, housing market correction, mortgage crisis, credit market disruption, financial market stress, liquidity support, bear stearns, lehman, debt crisis, sovereign stress,

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spreads widening, greek crisis, fragmentation, abenomics, reflation, provide liquidity, economic weakness, financial stress, flight to quality, downward pressure on prices, market turbulence, non-performing loans, public sector investment, peripheral countries, austerity, accommodative, low inflation, weak growth, room for manoeuvre, room for maneuver, output gap, soft landing, interbank market, uncertainty, headwinds, financial stability.

B. LLM Prompt

The LLM evaluates each speech using the prompt reproduced below, assigning a score from -1 (dovish) to +1 (hawkish). It also reports its confidence and short excerpts supporting the classification.

System prompt. You are a monetary policy text analyst. Score each speech on a hawk/dove scale from -1.0 (very dovish) to +1.0 (very hawkish). Return strict JSON with fields: raw_score, confidence, relevance, hawkish_excerpt, dovish_excerpt, central_bank, country. Confidence/relevance must be in [0,1]. central_bank should be the full name of the central bank (e.g. 'Federal Reserve', 'Bank of England'). country should be the ISO 3166-1 alpha-2 country code (e.g. 'US', 'GB', 'CN'). Scoring Guidelines: > 0.0 to +1.0 (Hawkish): The text suggests tightening bias, inflation concern, or restrictive policy stance. Closer to +1.0: Strong hawkish signals — aggressive tightening, urgent inflation warnings, or explicit rate hike guidance. Closer to 0.0: Mild hawkish lean — cautious language, data-dependent tightening, or moderate inflation vigilance. 0.0 (Neutral): The text presents a balanced policy stance, acknowledges both upside and downside risks equally, or expresses genuine uncertainty with no directional bias. -1.0 to < 0.0 (Dovish): The text suggests easing bias, growth concern, or accommodative policy stance. Closer to 0.0: Mild dovish lean — patience signaling, gradual easing hints, or downside risk acknowledgment. Closer to -1.0: Strong dovish signals — aggressive easing, explicit rate cut guidance, or urgent growth/employment support.

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1. The speeches were obtained from the Bank for International Settlements' Central Bankers' Speeches database, available at https://www.bis.org/cbspeeches/download.htm . Return to text
2. The complete list of hawkish and dovish terms used by the lexicon model is provided in the Appendix. Return to text
3. We accessed the model through Anthropic's API and set it to minimize variation when evaluating the same speech more than once. The exact instructions supplied to the model are reproduced in the Appendix. Return to text

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Processing math: 100%